

Rules of Inference :-1. Modus ponens :- (rule of detachment)

$$\begin{array}{l} p \rightarrow q \\ p \\ \hline \therefore q \end{array}$$

$\hookrightarrow (p \rightarrow q) \wedge p \rightarrow q$ is a tautology.

eg. P_1 : If it is snowing, then it is cold outside.
 P_2 : It is snowing. $\rightarrow p$
 $\hline \therefore q$: It is cold outside.

2. Modus tollens (Rule of Contraposition) :-

$$\begin{array}{l} p \rightarrow q \\ \neg q \\ \hline \therefore \neg p \end{array} \quad \left. \vphantom{\begin{array}{l} p \rightarrow q \\ \neg q \\ \hline \therefore \neg p \end{array}} \right\} \text{valid argument.}$$

$\hookrightarrow ((p \rightarrow q) \wedge (\neg q)) \rightarrow \neg p$ is a tautology.

	p	q	$\neg p$	$\neg q$	$(p \rightarrow q)$ ^(x)	$(x \wedge \neg q)$ ^(y)	$y \rightarrow \neg p$
x	T	F	F	T	F		
	<u>F</u>	<u>F</u>	T	T	T	T	<u>T</u>

$\hookrightarrow$ It's a tautology.

$$\begin{aligned} \neg q &= T \\ q &= F \\ p \rightarrow q &\equiv T \end{aligned}$$

eg. P_1 : If the car is out of gas, then it won't start.

P_2 : The car is starting.

$\therefore Q$: The car is not out of gas.

(3) Hypothetical Syllogism :- (Transitivity)

$$\begin{array}{l} p \rightarrow q \\ q \rightarrow r \\ \hline \therefore p \rightarrow r \end{array}$$

} valid argument.

↳ $((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$ is a tautology.

eg. P_1 : If I study hard, then I will pass the exam.

P_2 : If I pass the exam, then I will get a scholarship.

$\therefore Q$: If I study hard, then I will get a scholarship.

(4) Disjunctive Syllogism :-

$$\begin{array}{l} p \vee q \\ \neg p \\ \hline \therefore q \\ \text{—} \end{array}$$

↳ $(p \vee q) \wedge (\neg p) \rightarrow q$ is a tautology.

(5) Conjunctive Syllogism :-

$$\begin{array}{l} p \vee q \\ \neg q \\ \hline \therefore p \end{array}$$

$$\begin{array}{l} \text{F} \\ \text{p} \vee \text{q} \equiv \text{T} \\ \neg \text{p} \equiv \text{T}, \text{p} = \text{F} \end{array}$$

p	q
F	T

⑥ Addition :-

$$\frac{p}{\therefore p \vee q}$$

$$p \equiv T \\ \therefore T \vee q \equiv \textcircled{T}$$

⑦ Simplification :-

$$\frac{p \wedge q}{\therefore p}$$

$$\begin{array}{c} T \quad T \\ \textcircled{p} \wedge \textcircled{q} \equiv T \\ \therefore \boxed{p \equiv T} \end{array}$$

⑧ Conjunction :-

$$\frac{\begin{array}{c} p \\ q \end{array}}{\therefore p \wedge q}$$

$$\begin{array}{c} p \rightarrow T \\ q \rightarrow T \\ \therefore \boxed{p \wedge q \Rightarrow T} \end{array}$$

Q9. Resolution :-

$$\begin{array}{l} p \vee q \\ \neg p \vee r \\ \hline \therefore [q \vee r] \end{array}$$

p	q	r	$\neg p$	(p $\vee$ q)	($\neg p \vee r$)	$p \wedge q$	q $\vee$ r	$z \rightarrow t$
T	T	T	F	T	T	T	T	T
T	F	T	F	T	T	T	T	T
F	T	T	T	T	T	T	T	T
F	T	F	T	T	T	T	T	T

$\hookrightarrow (p \vee q) \wedge (\neg p \vee r) \rightarrow (q \vee r)$ is a tautology.

eg. P_1 : $\overset{p}{\text{Either the dog is outside}}, \text{ or } \overset{q}{\text{the door is open.}}$
 P_2 : $\text{Either } \overset{\neg q}{\text{the door is not open}}, \text{ or } \overset{r}{\text{someone is home.}}$
 $\therefore Q$: $\overset{p}{\text{Either the dog is outside}}, \text{ or } \overset{r}{\text{someone is home.}}$

$$\begin{array}{l} P_1: (p \vee q) \equiv q \vee p \\ P_2: \neg q \vee r \equiv \neg q \vee r \\ \hline [p \vee r] = \end{array}$$

10. Dilemma :-

$$\begin{array}{l} p \vee q \\ p \rightarrow r \\ q \rightarrow r \\ \hline \therefore r \end{array}$$

$\hookrightarrow ((p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)) \rightarrow r$ is a tautology. (H.W.)

11. Constructive dilemma :-

$$\begin{array}{l} p \vee q \\ p \rightarrow r \\ q \rightarrow s \\ \hline \therefore r \vee s \end{array}$$

$\hookrightarrow ((p \vee q) \wedge (\neg p \vee r) \wedge (\neg q \vee s)) \rightarrow r \vee s$ is a tautology. (H.W.)

(12.) Destructive dilemma :-

$$\begin{array}{l} p \rightarrow r \\ q \rightarrow s \\ \neg r \rightarrow \neg s \\ \hline \therefore \neg p \vee \neg q \end{array} \quad \left. \vphantom{\begin{array}{l} p \rightarrow r \\ q \rightarrow s \\ \neg r \rightarrow \neg s \end{array}} \right\} \text{valid arg.}$$

$\hookrightarrow ((\neg p \vee r) \wedge (\neg q \vee s) \wedge (r \vee \neg s)) \rightarrow (\neg p \vee \neg q)$ is a tautology. ^{H.W.}

Q. Stmt: It is below freezing & raining now. Therefore, it is below freezing now.

- a) Addition
- b) Simplification
- c) Conjunction
- d) Modus ponens.

Valid / Invalid?

$$\left\{ \begin{array}{l} \frac{P \wedge Q}{\therefore P} \end{array} \right. \underline{\text{valid.}}$$

Q. Are these arguments valid or not.

a) $\frac{Q}{\therefore P \rightarrow Q}$ valid.

$\boxed{P}$	$\sim Q$	Q
$\underline{\underline{T}}$	T	$\boxed{F}$ x

b) $\frac{\sim (P \rightarrow Q)}{\therefore P \sim}$ valid.

$$\begin{aligned} \boxed{\sim (P \rightarrow Q)} &\equiv \sim (\sim P \vee Q) \\ &\equiv \boxed{P \wedge \sim Q} \\ &\quad \downarrow \quad \downarrow \\ &\quad T \quad T \rightarrow \boxed{Q \equiv F} \end{aligned}$$

$$\boxed{P \wedge \sim Q}$$

c) $\frac{\sim (P \rightarrow Q)}{\therefore Q \times}$ Invalid.

$\frac{\sim (P \rightarrow Q)}{\therefore \sim Q}$ valid.

Q. P_1 : It is not sunny this afternoon and it is colder than yesterday.

P_2 : We will go swimming only if it is sunny.

P_3 : If we do not go swimming, then we will take a trip.

P_4 : If we take a trip, then we will be home by sunset.

$\therefore Q$: We will be home by sunset.

valid or invalid?

Sol. $\neg p$ q

P_1 : It is not sunny this afternoon and it is colder than yesterday.

P_2 : We will go swimming only if it is sunny. $\rightarrow p$

P_3 : If we do not go swimming, then we will take a trip. (S)

P_4 : If we take a trip, then we will be home by sunset. (T)

$\therefore Q$: We will be home by sunset.

Argument form

$$P_1: \neg p \wedge q$$

$$P_2: r \rightarrow p$$

$$P_3: \neg r \rightarrow s$$

$$P_4: s \rightarrow t$$

$$\therefore Q: t.$$

$$\neg p \wedge q \equiv T \rightarrow \text{Simplification.}$$

$$\therefore \neg p \equiv T.$$

$$\begin{array}{l} r \rightarrow p \\ \neg p \end{array} \left. \vphantom{\begin{array}{l} r \rightarrow p \\ \neg p \end{array}} \right\} \text{modus tollens.}$$

$$\therefore \neg r \quad (T)$$

$$\begin{array}{l} P_3: \neg r \rightarrow s \\ \neg r \end{array} \left. \vphantom{\begin{array}{l} \neg r \rightarrow s \\ \neg r \end{array}} \right\} \text{modus ponens.}$$

$$\therefore s \quad (T)$$

$$\begin{array}{l} P_4: s \rightarrow t \\ s \end{array} \left. \vphantom{\begin{array}{l} s \rightarrow t \\ s \end{array}} \right\} \text{modus ponens.}$$

$$\therefore t \equiv T \quad \therefore \text{valid argument.}$$

H.W.

P_1 : If you send me an e-mail message, then I will finish writing the program.

P_2 : If you do not send me an e-mail message, then I will go to sleep early.

P_3 : If I go to sleep early, then I will wake up feeling refreshed.

$\therefore Q$: If I do not finish writing the program, then I will wake up feeling refreshed.

valid or Invalid ?